

PRIVATE CREDENTIAL LAYER · BUILT ON AZTEC

Magna

Verify once. Prove anywhere. Reveal nothing.

A real-world identity proof — like a zkPassport scan — becomes a private credential that apps can trust without ever seeing the underlying data.

1 • The Problem

Identity online is a data-hoarding model

Every app that needs to know "are you real / over 18 / from an allowed country" does the same thing:

- **Collects your raw documents** — passport scan, selfie, date of birth.
- **Stores that PII** on its own servers, forever.
- **Re-verifies from scratch** the next time, at the next app, and the next.

The result: hundreds of copies of your most sensitive data, sitting in hundreds of breachable databases — and a slow, repeated onboarding tax on every user.

That broken model is expensive — for everyone

The cost today	
Avg. KYC review, corporate client	\$2,598
Avg. bank's annual KYC spend	\$60M+ (majors \$500M+)
Avg. time to complete a KYC review	~95 days
Repetitive KYC, EU institutions / yr	€5B+
Avg. cost of a single data breach	~\$4.4M

And it costs **customers** too: **60–80%** of users abandon digital onboarding, and **~70%** drop off any flow that takes longer than **3 minutes**.

<footer>Sources: Fenergo / ShareRing 2026 · Sumsb · IBM Cost of a Data Breach 2025 · Jumio · Zyphe. See appendix.</footer>

2 • The Solution

Magna: prove the fact, never share the data

Magna turns a one-time real-world proof into a **private, reusable credential** living on a private blockchain (Aztec).

- The app learns **only what it needs** — "verified ✓", "over 18", "allowed nationality."
- It **never receives** your name, document number, photo, or date of birth.
- **No one** — not the app, not Magna — stores your raw passport data.

Verify once → reuse everywhere

How it works — in three steps

1 · Prove

You scan your passport with **zkPassport** on your own phone. A zero-knowledge proof is generated **on-device** — the raw chip data never leaves it.

2 · Commit

Magna verifies the proof and issues a **private credential** on Aztec. Only opaque cryptographic **commitments** are stored — never the underlying values.

3 · Reuse

Any integrating app asks a question ("over 18?", "allowed country?"). Magna answers **true/false** with a fresh proof — **gaslessly, in seconds**.

What makes it different: PII-blind by construction

This isn't a privacy *policy* — it's privacy enforced by **math**.

- The credential stores a **commitment** = a one-way hash of (value, random blind).
- The real values + blinds — the "witness" — **stay on the user's device**.
- The backend verifies proofs and issues credentials having **seen zero raw PII**.

Even Magna's own servers cannot reveal a user's nationality or birth date. There is no honeypot to breach, no database to subpoena, no PII to leak.

Credentials that behave like real-world ID

A passport isn't single-use — and neither is a Magna credential. Three properties competitors rarely combine:

Property	What it means
Reusable	Verify once; prove to unlimited apps with no re-scan.
Recoverable	Lose your device? A deterministic recovery path restores control — without a central account to hack.
Revocable	Expired or stolen passport pauses/kills the credential; renewal rotates it cleanly.

3 · The Product

"Login with Magna" — identity as one button

For apps, integration is as simple as "Sign in with Google" — but private.

```
const result = await magna.login({
  credentialType: "Passport",
  constraints: [ ageGte(18), countryNeq("USA") ],
});
if (result.verified) unlockApp();
```

- Drop-in SDK; **three steps** to integrate.
- The app gets back `{ verified }` + a signed assertion — **no keys, notes, claims, or identifiers**.
- Verification is **gasless** for the user.

Architecture at a glance

```
flowchart LR
    U["User device<br/>passport scan + private keys"] -->|ZK proof only| API["Magna<br/>Verification API"]
    API -->|issues credential| AZ["Aztec private L2<br/>commitments only"]
    APP["Integrating app<br/>'Login with Magna'"] -->|asks a question| U
    U -->|verified / true-false| APP
```

- **On device:** raw data + secrets. **On chain:** opaque commitments. **At the app:** only the answer.
- Built on **Aztec**, a privacy-first L2 — private state is the default, not an add-on.

<footer>Mermaid renders on GitHub/Notion/Obsidian/mermaid.live and in Marp with the Mermaid plugin; export to an image for PowerPoint/Keynote.</footer>

4 · The Opportunity

A large market, with a hyper-growth core

Market	2025/26	Forecast	CAGR
Digital identity solutions (TAM)	~\$51B	~\$205B (2034)	~18%
Identity verification (SAM)	~\$15–17B	~\$32B (2030)	~17%
Decentralized / self-sovereign ID (SOM)	~\$3–7B	~\$38–75B (2030)	66–85%

Magna sits in the **fastest-growing slice** — decentralized, privacy-preserving identity — riding the much larger identity-verification spend as it migrates to reusable, user-controlled credentials.

<footer>Sources: Fortune Business Insights · TBRC · Technavio · Mordor Intelligence · Grand View Research. Ranges reflect differing report scopes. See appendix.</footer>

Why now

Three curves are crossing at the same moment:

- **Regulation** — GDPR/CCPA data-minimization and EU digital-identity mandates turn "don't hoard PII" from nice-to-have into **compliance necessity**.
- **Technology** — zero-knowledge proving and **on-device identity proofing** are now production-ready and cheap enough to run on a phone.
- **Demand** — breach fatigue and KYC drop-off push both users and businesses toward **verify-once, reveal-nothing** credentials.

Business model

Magna monetizes **verification**, not data.

- **Per-verify metering** — integrating apps pay per successful proof through registered gateways.
- **Sponsored / gasless flows** — companies sponsor their users' verifications (covered via on-chain rights), removing all user friction.
- **Reusable-credential economics** — repeat verifications resolve at a fraction of a full KYC check, the saving Magna shares with integrators.

Aligned incentive: we make money when apps successfully verify users — and we hold none of the liability that comes with storing PII.

5 · Status & Roadmap

What's already built

This is not a concept — the full stack is implemented and running locally end-to-end.

- **Protocol contracts** on Aztec (issuance, verify, recovery, renewal, sponsorship).
- **TypeScript SDK** — the thin "Login with Magna" connector for any dApp.
- **Off-chain verification API** for zkPassport proofs + credential issuance.
- **Two reference apps** — a Management dApp and a Reference (integration) dApp.
- **PII-blind A1 flow live** — only Poseidon2 commitments leave the device.

Roadmap

Now	Next	Later
PII-blind passport credentials (live)	WebAuthn passkey accounts (keys never touch JS)	Additional credential types beyond passport
Reusable verify + recovery	Production security audit	Broader issuer/verifier network
Single-tenant gateways	Multi-tenant consumer gateways	Salted OPRF nullifiers for unlinkable, Sybil-resistant identity

Why Magna wins

	Centralized IDV (Jumio, Onfido...)	Most SSI wallets	
Stores raw PII	Yes (honeypot)	Sometimes	Never
Reusable across apps	No	Partial	Yes
Recoverable + revocable	N/A	Rare	Yes
Gasless for the user	N/A	No	Yes
Developer drop-in	Varies	Hard	One call

The future of identity isn't *more* data.

It's proof without exposure.

Magna makes "verify once, prove anywhere, reveal nothing" the default — for every app, every user, every time.

VERIFY ONCE · PROVE ANYWHERE · REVEAL NOTHING

Appendix — sources

- **Digital identity solutions market** — Fortune Business Insights (2025: \$43B → 2034: \$205B, 18.9% CAGR); Precedence Research (\$55.7B 2026 → \$231B 2035).
- **Identity verification market** — The Business Research Co. / giiresearch (\$14.8B 2025 → \$32.5B 2030, ~17% CAGR); Technavio (+\$25.7B 2026–2030, 21.9% CAGR).
- **Decentralized / self-sovereign identity** — Mordor Intelligence (\$3.25B 2025 → \$65.6B 2030, 82.4% CAGR); Grand View Research (\$1.9B 2024 → \$38.1B 2030, 66.8%); TBRC (\$3.78B 2025 → \$74.9B 2030, ~82%).
- **KYC cost** — Fenargo via ShareRing 2026 (\$2,598 / corporate client; bank avg \$60M/yr; ~95-day review).
- **Repetitive KYC cost** — Sumsb (EU institutions €5B+/yr; drop-off >50%).
- **Onboarding abandonment** — Jumio (60–80%; 68% of banking apps); Zyphe (70% abandon flows >3 min).
- **Reusable-credential savings** — Mordor via ShareRing (30–50% onboarding cost cut; up to 60% repeat-check reduction); Zyphe / Sumsb (~70% higher completion; ~30% higher conversion).